

22UCV531N	Air Pollution and Control	Credits: 03
L:T:P - 3 : 0 : 0		CIE Marks: 50
Total Hours/Week: 3		SEE Marks: 50

UNIT-I		10 Hrs.
Introduction: Definition, Sources, classification and characterization of air pollutants. Effects of air pollution on health, vegetation & materials. Atmosphere and water bodies, Photo-chemical Smog, .		
UNIT-II		10 Hrs.
Meteorology: Temperature lapse rate & stability, wind velocity & turbulence, plume behavior, measurement of meteorological variables, wind rose diagrams, Plume Rise, estimation of effective stack height and mixing depths. Development of air quality models-Gaussian dispersion model and Numerical problems.		
UNIT-III		10Hrs.
Sampling: Sampling of particulate and gaseous pollutants (Stack, Ambient & indoor air pollution), Monitoring and analysis of air pollutants (PM _{2.5} , PM ₁₀ , SO _x , NO _x , CO, NH ₃) and Air pollution emission standards and Numerical problems		
UNIT-IV		10 Hrs.
Control Techniques: Air pollution control devices, equipment and their design. Particulate matter and gaseous pollutants- settling chambers, cyclone separators, scrubbers, filters & ESP. Including Numerical problems. Indoor air quality- sources, types and control of air pollutants		
Reference Books *		
1. M. N. Rao and H V N Rao, "Air pollution", Tata Mc-G raw Hill Publication. 2. H. C. Perkins, "Air pollution". Tata McGraw Hill Publication 3. Mackenzie Davis and David Cornwell, " Introduction to Environmental Engineering" McGraw-Hill Co. Noel De Nevers, "Air Pollution Control Engineering" , Waveland Pr Inc. Anjaneyulu Y, "Text book of Air Pollution and Control Technologies", Allied Publishers		
Course Outcomes		
After studying this course, students will be able to: 1. Identify the major sources of air pollution and understand their effects on health and environment. 2. Evaluate the dispersion of air pollutants in the atmosphere and to develop air quality models. 3. Ascertain and evaluate sampling techniques for atmospheric and stack pollutants. 4. Choose and design control techniques for particulate and gaseous emissions		

Course Outcomes	Programme Outcomes												Programme Specific Outcomes		
	1	2	3	4	5	6	7	8	9	10	11	12	PSO1	PSO2	PSO3
CO 1	1	2	1								1	1	2	1	
CO 2	1	1	2								2	1	2	1	
CO 3	1	2	1								1	1	2	1	
CO 4	1	1	1								2	1	2	1	

Assignment Questions

Unit 1

1. **Define air pollution** and explain the criteria used to identify a substance as an air pollutant.
2. **Classify air pollutants** based on origin, physical state, and chemical composition, with suitable examples.
3. Explain the **natural and anthropogenic sources of air pollution** and discuss their relative contribution.
4. What is meant by **primary and secondary pollutants**? Give at least two examples of each.
5. Describe the **characterization of air pollutants** in terms of particle size, chemical reactivity, and persistence in the environment.
6. Discuss the **effects of air pollution on human health**, citing examples of diseases caused by particulate matter, sulfur dioxide, carbon monoxide, and ozone.
7. Explain how **air pollution affects vegetation** and agricultural productivity. Illustrate with at least two case studies or examples.
8. Discuss the **impact of air pollutants on materials** such as metals, building stones, and cultural monuments.
9. Explain the **interaction of air pollutants with the atmosphere and water bodies**. How do these interactions contribute to problems such as acid rain?
10. What is **photochemical smog**? Explain its formation mechanism, major components, and harmful effects on health and environment.

Unit 2

1. Define **temperature lapse rate** and explain the difference between *environmental lapse rate*, *adiabatic lapse rate*, and *inversion*.
2. What is meant by **atmospheric stability**? Discuss its role in the vertical and horizontal dispersion of air pollutants.
3. Explain the importance of **wind velocity** in the transport and dilution of air pollutants.
4. Define **turbulence** and explain how mechanical and thermal turbulence influence dispersion.
5. Illustrate and explain the different **plume behaviors** (looping, coning, fanning, fumigation, lofting, and trapping).
6. Discuss the **methods of measuring meteorological variables** such as wind speed, wind direction, humidity, solar radiation, and atmospheric pressure.
7. What is a **wind rose diagram**? Explain its significance in determining the impact of industrial emissions on surrounding areas.
8. Define **plume rise**. What are the factors that control plume rise in the atmosphere?
9. Explain the concept of **effective stack height**. Why is it more important than physical stack height for pollution dispersion studies?
10. What is meant by **mixing depth**? Discuss how it affects the dilution of air pollutants.
11. Describe the **development of air quality models** and their applications in environmental management.
12. Explain the **Gaussian dispersion model** with its assumptions, limitations, and practical applications.